Advanced Python for Data Science
DS-GA 1019

Homework 01

Due date: Feb 13, 2026, 11:00PM

Instructions for submitting solutions for assignment 1:

- Please submit only one zip file to NYU Classes, which contains **all your python solutions** and **one pdf file** with all of your answers.
- The submitted zip-file name should be as **"spring2026_sol01_*nyuid*.zip"**
- **e.g.** If your nyuid is "ab1234", the submission is "spring2026_sol01_ab1234.zip"

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Problem 1. (30pt)

Write a function f(n) to calculate the number of ways of representing n as a sum of 1, 2, and 5, where the order of summands is important. For example:

---

n = 1
1 = 1
f(1) = 1

---

n = 2
2 = 1 + 1
2 = 2
f(2) = 2

---

n = 3
3 = 1 + 1 + 1
3 = 1 + 2
3 = 2 + 1
f(3) = 3

---

n = 5
5 = 1 + 1 + 1 + 1 + 1
5 = 1 + 1 + 1 + 2
5 = 1 + 1 + 2 + 1
5 = 1 + 2 + 1 + 1
5 = 2 + 1 + 1 + 1
5 = 2 + 2 + 1
5 = 2 + 1 + 2
5 = 1 + 2 + 2
5 = 5
f(5) = 9

(10pt) Write a recursive solution f_rec() without memoization and time it for $n = 10, 25$.

(10pt) Write a recursive solution with memoization or a wrapper function f_memo() and time it for $n = 10, 25, 50, 100$.

(10pt) Write an iterative solution f_it() and time it for $n = 10, 25, 50, 100$.

In [1]:
```
def f_rec(n):
    if n < 0:
        return 0
```

```
    elif n == 0:
        return 1
    else:
        return f_rec(n-1) + f_rec(n-2) + f_rec(n-5)
```

In [2]:
```
%timeit -r 10 -n 20 f_rec(10)
%timeit -r 10 -n 20 f_rec(25)
```

```
56.4 µs ± 5.22 µs per loop (mean ± std. dev. of 10 runs, 20 loops each)
117 ms ± 3.61 ms per loop (mean ± std. dev. of 10 runs, 20 loops each)
```

In [3]:
```
from functools import lru_cache

@lru_cache(maxsize=128)
def f_memo(n):
    if n < 0:
        return 0
    elif n == 0:
        return 1
    else:
        return f_memo(n-1) + f_memo(n-2) + f_memo(n-5)
```

In [4]:
```
%timeit -r 10 -n 20 f_memo(10)
%timeit -r 10 -n 20 f_memo(25)
%timeit -r 10 -n 20 f_memo(50)
%timeit -r 10 -n 20 f_memo(100)
```

```
67.3 ns ± 78.3 ns per loop (mean ± std. dev. of 10 runs, 20 loops each)
69.4 ns ± 80.4 ns per loop (mean ± std. dev. of 10 runs, 20 loops each)
81.9 ns ± 123 ns per loop (mean ± std. dev. of 10 runs, 20 loops each)
121 ns ± 240 ns per loop (mean ± std. dev. of 10 runs, 20 loops each)
```

In [5]:
```
def f_it(n):
    dp = [0] * (n + 2)
    dp[0] = 0
    dp[1] = 1

    for i in range(2, n + 2):
        dp[i] = dp[i-1] + dp[max(0, i-2)] + dp[max(0, i-5)]

    return dp[n+1]
```

In [6]:
```
%timeit -r 10 -n 20 f_it(10)
%timeit -r 10 -n 20 f_it(25)
%timeit -r 10 -n 20 f_it(50)
%timeit -r 10 -n 20 f_it(100)
```

```
3.51 µs ± 1.24 µs per loop (mean ± std. dev. of 10 runs, 20 loops each)
6.72 µs ± 279 ns per loop (mean ± std. dev. of 10 runs, 20 loops each)
13.7 µs ± 1.32 µs per loop (mean ± std. dev. of 10 runs, 20 loops each)
26.7 µs ± 1.45 µs per loop (mean ± std. dev. of 10 runs, 20 loops each)
```

Problem 2. (70%)

An astrophysicist colleague was recently complaining about how long it was taking to run an N-body simulation. “It’s really just a simple calculation, and I’m only simulating four planets, but it takes nearly a minute and a half to run one simulation. I really need it done in under 30 seconds.” You kindly offer to take a look at code to see if it is possible to speed it up. Your colleague provides you with a link to the source: https://nyu-cds.github.io/courses/code/nbody.py

Although your colleague said the code was simple, it is still fairly complex, so you decide to tackle the problem in stages. A first scan of the code reveals a number of potential areas that could be improved. These include:

- Reducing function call overhead
- Using alternatives to membership testing of lists
- Using local rather than global variables
- Using data aggregation to reduce loop overheads.

As you’re a cautious programmer, you decide to address each of these in turn. This will ensure that it is possible to check the program is still working correctly after each change, and to assess the performance improvement that the change achieved. You are also aware that the program has to be maintained by others in the future, so you want to make sure that the changes do not make this more difficult, especially if the performance improvement is only minor.

For each of these areas, create a new version of nbody.py, call them nbody_1.py, nbody_2.py, nbody_3.py, nbody_4.py. Finally, create another file called nbody_opt.py that contains all of the optimizations you made.

Finally, generate a .pdf file with the four logs produced by the command in shell:

for f in nbody_*.py; do python -m cProfile -s cumulative $f; done

How much speedup do you get (time_original/time_optimized)?

In [ ]:
```
! for f in assignments/hw1/nbody_*.py; do python -m cProfile -s cumulative $f; done
```

0.5316285081085336
0.5341722321050918
0.5350125560137375
0.535431017191156
0.5356814963634876
0.5358482173279006
0.5359671675805173
0.536056302957533
0.5361255832078345
0.5361809768305218
0.5362262781976399
0.5362640148219056
0.5362959353162482
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0.5363469876776918
0.5363677203785557
0.5363860103622478
0.5364022652780793
0.5364168068534976
0.5364298923978392
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0.5364524904692427
0.5364623140256721
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0.5364796009606672
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0.5364943242976354
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0.5365127250908598
0.5365180665446689
0.5365230738639242
0.5365277774484875
0.5365322041199077
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0.5365679042372806
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0.5365769096557202
0.5365789357384417
0.5365808823190212
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0.5365845549859037
0.5365862892432378
0.5365879604005354
0.5365895718407794
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0.5365954801756647
0.5365968361258082
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0.536601842504845
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0.5366092645652313
0.5366102086120345
0.5366111271331543
0.5366120211494431
0.536612891628583
0.5366137394879766
0.5366145655987793
0.5366153707868533
0.5366161558365519
0.536616921494889
0.5366176684715074
0.5366183974407764
0.5366191090467326

```
0.5366198039026356
0.5366204825923996
0.536621145673687
0.5366217936797877
0.5366224271194284
0.5366230464774651
0.5366236522183147
0.5366242447853409
0.5366248246045144
0.5366253920837349
0.5366259476111767
0.5366264915616991
0.5366270242922768
0.5366275461465295
0.5366280574558655
0.5366285585358422
         36003706 function calls in 19.164 seconds

   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000   19.164   19.164 {built-in method builtins.exec}
        1    0.000    0.000   19.164   19.164 nbody_1.py:1(<module>)
        1    0.297    0.297   19.164   19.164 nbody_1.py:151(nbody)
  2000000   17.721    0.000   18.863    0.000 nbody_1.py:67(advance)
 20002000    0.682    0.000    0.682    0.000 {method 'append' of 'list' objects}
 14001401    0.460    0.000    0.460    0.000 {method 'keys' of 'dict' objects}
      100    0.002    0.000    0.002    0.000 {built-in method builtins.print}
      200    0.002    0.000    0.002    0.000 nbody_1.py:107(report_energy)
        1    0.000    0.000    0.000    0.000 nbody_1.py:134(offset_momentum)
        1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}


-0.16908926275527172
-0.1690524049239141
-0.16901474404530115
-0.16902307738080738
-0.1690798593916716
-0.1690837851340391
-0.1690461237029509
-0.1690141632287673
-0.1690277860596755
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-0.16908371256964286
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-0.16901196403207439
-0.16903317301080864
-0.16908783999484456
-0.16908019793585344
-0.16904027123722912
-0.16900666759845165
-0.1690385994000913
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-0.16904872400310705
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-0.16901235909925366
-0.16901736033369455
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-0.16909035534676262
-0.16904838452880422
-0.16900663034076266
-0.16902335367423574
-0.16909124149510837
-0.16908618459859356
-0.16904015517545168

-0.16900291923555205
-0.16903287494400335
-0.16909558331070956
-0.16908055117120765
-0.16903390308620184
-0.16900274590721814
-0.16904459300211724
-0.16909854774651836
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-0.16902883385621698
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-0.16901176533650106
-0.16901316541886924
-0.16908223578125162
-0.1690918842838697
-0.1690491567289713
-0.1690084483625472
-0.16902057135969534
-0.16908716660200046
-0.16908658831414558
-0.16904412116934314
-0.16900740673825185
-0.16902932050670785
-0.16909180276278374
-0.16908292959405574
-0.16903991544261052
-0.16900522540601676
-0.16903656773659115
-0.16909410318888132

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-0.169078466211733
-0.16903274720873912
-0.16900284987920383
-0.1690462807510843
-0.1690971767866576
-0.16907293588692712
-0.16902628585294033
         326007706 function calls in 63.341 seconds

   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000   63.341   63.341 {built-in method builtins.exec}
        1    0.000    0.000   63.341   63.341 nbody_2.py:1(<module>)
        1    0.314    0.314   63.341   63.341 nbody_2.py:129(nbody)
  2000000   13.598    0.000   63.023    0.000 nbody_2.py:67(advance)
 20000000   16.313    0.000   44.716    0.000 nbody_2.py:54(update_vs)
120000000   12.730    0.000   28.403    0.000 nbody_2.py:47(compute_b)
120000000   15.673    0.000   15.673    0.000 nbody_2.py:51(compute_mag)
 10000000    1.736    0.000    1.736    0.000 nbody_2.py:62(update_rs)
 20002000    1.530    0.000    1.530    0.000 nbody_2.py:44(compute_deltas)
 20002000    0.853    0.000    0.853    0.000 {method 'add' of 'set' objects}
 14001401    0.591    0.000    0.591    0.000 {method 'keys' of 'dict' objects}
      200    0.002    0.000    0.002    0.000 nbody_2.py:90(report_energy)
      100    0.002    0.000    0.002    0.000 {built-in method builtins.print}
     2000    0.000    0.000    0.000    0.000 nbody_2.py:87(compute_energy)
        1    0.000    0.000    0.000    0.000 nbody_2.py:112(offset_momentum)
        1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}


-0.16908926275527172
-0.1690524049239141
-0.16901474404530115
-0.16902307738080738
-0.1690798593916716
-0.1690837851340391
-0.1690461237029509
-0.1690141632287673
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-0.16908371256964286
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-0.16904474158017624
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-0.16908716660200046
-0.16908658831414558
-0.16904412116934314
-0.16900740673825185
-0.16902932050670785
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-0.16903991544261052
-0.16900522540601676
-0.16903656773659115
-0.16909410318888132
-0.169078466211733
-0.16903274720873912
-0.16900284987920383

```
-0.1690462807510843
-0.1690971767866576
-0.16907293588692712
-0.16902628585294033
         326007706 function calls in 64.656 seconds

   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000   64.656   64.656 {built-in method builtins.exec}
        1    0.000    0.000   64.656   64.656 nbody_3.py:1(<module>)
        1    0.322    0.322   64.656   64.656 nbody_3.py:135(nbody)
  2000000   14.490    0.000   64.330    0.000 nbody_3.py:67(advance)
 20000000   16.379    0.000   45.074    0.000 nbody_3.py:54(update_vs)
120000000   12.804    0.000   28.695    0.000 nbody_3.py:47(compute_b)
120000000   15.891    0.000   15.891    0.000 nbody_3.py:51(compute_mag)
 10000000    1.745    0.000    1.745    0.000 nbody_3.py:62(update_rs)
 20002000    1.552    0.000    1.552    0.000 nbody_3.py:44(compute_deltas)
 20002000    0.863    0.000    0.863    0.000 {method 'append' of 'list' objects}
 14001401    0.606    0.000    0.606    0.000 {method 'keys' of 'dict' objects}
      200    0.002    0.000    0.002    0.000 nbody_3.py:93(report_energy)
      100    0.002    0.000    0.002    0.000 {built-in method builtins.print}
     2000    0.000    0.000    0.000    0.000 nbody_3.py:90(compute_energy)
        1    0.000    0.000    0.000    0.000 nbody_3.py:117(offset_momentum)
        1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}


-0.16908926275527172
-0.1690524049239141
-0.16901474404530115
-0.16902307738080738
-0.1690798593916716
-0.1690837851340391
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```

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-0.16901736033369455
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-0.16909035534676262
-0.16904838452880422
-0.16900663034076266
-0.16902335367423574
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-0.16908618459859356
-0.16904015517545168
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-0.16903390308620184
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-0.16900287426063337
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-0.16909778695827116
-0.1690688542355601
-0.16902288517135491
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-0.16907413362652127
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-0.16905715253841647
-0.16901176533650106
-0.16901316541886924
-0.16908223578125162
-0.1690918842838697
-0.1690491567289713
-0.1690084483625472
-0.16902057135969534
-0.16908716660200046
-0.16908658831414558
-0.16904412116934314
-0.16900740673825185
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-0.16903991544261052
-0.16900522540601676
-0.16903656773659115
-0.16909410318888132
-0.169078466211733
-0.16903274720873912
-0.16900284987920383
-0.1690462807510843
-0.1690971767866576
-0.16907293588692712

```
-0.16902628585294033
         316006706 function calls in 61.308 seconds

   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000   61.308   61.308 {built-in method builtins.exec}
        1    0.000    0.000   61.308   61.308 nbody_4.py:1(<module>)
        1    0.311    0.311   61.308   61.308 nbody_4.py:125(nbody)
  2000000   11.814    0.000   60.993    0.000 nbody_4.py:69(advance)
 20000000   16.247    0.000   44.921    0.000 nbody_4.py:56(update_vs)
120000000   12.840    0.000   28.674    0.000 nbody_4.py:49(compute_b)
120000000   15.834    0.000   15.834    0.000 nbody_4.py:53(compute_mag)
 10000000    1.724    0.000    1.724    0.000 nbody_4.py:64(update_rs)
 20002000    1.513    0.000    1.513    0.000 nbody_4.py:46(compute_deltas)
 20002000    0.822    0.000    0.822    0.000 {method 'append' of 'list' objects}
  4000401    0.199    0.000    0.199    0.000 {method 'keys' of 'dict' objects}
      100    0.002    0.000    0.002    0.000 {built-in method builtins.print}
      200    0.001    0.000    0.002    0.000 nbody_4.py:89(report_energy)
     2000    0.000    0.000    0.000    0.000 nbody_4.py:86(compute_energy)
        1    0.000    0.000    0.000    0.000 nbody_4.py:108(offset_momentum)
        1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}


0.5316285081085336
0.5341722321050918
0.5350125560137375
0.535431017191156
0.5356814963634876
0.5358482173279006
0.5359671675805173
0.536056302957533
0.5361255832078345
0.5361809768305218
0.5362262781976399
0.5362640148219056
0.5362959353162482
0.5363232879559959
0.5363469876776918
0.5363677203785557
0.5363860103622478
0.5364022652780793
```

0.5364168068534976
0.5364298923978392
0.5364417301557046
0.5364524904692427
0.5364623140256721
0.5364713180415462
0.5364796009606672
0.5364872460675782
0.5364943242976354
0.5365008964433411
0.5365070149050816
0.5365127250908598
0.5365180665446689
0.5365230738639242
0.5365277774484875
0.5365322041199077
0.5365363776320206
0.5365403190967136
0.5365440473433044
0.5365475792159419
0.5365509298321379
0.5365541127942481
0.5365571403787278
0.5365600236913958
0.5365627728032861
0.5365653968714682
0.5365679042372806
0.5365703025152019
0.5365725986737719
0.5365747990989369
0.5365769096557202
0.5365789357384417
0.5365808823190212
0.5365827539870915
0.5365845549859037
0.5365862892432378
0.5365879604005354
0.5365895718407794
0.5365911267084259
0.5365926279307529
0.5365940782365957
0.5365954801756647

```
0.5365968361258082
0.5365981483127226
0.5365994188215135
0.5366006496068751
0.536601842504845
0.5366029992366045
0.5366041214222812
0.5366052105862976
0.5366062681652415
0.5366072955145903
0.5366082939122084
0.5366092645652313
0.5366102086120345
0.5366111271331543
0.5366120211494431
0.536612891628583
0.5366137394879766
0.5366145655987793
0.5366153707868533
0.5366161558365519
0.536616921494889
0.5366176684715074
0.5366183974407764
0.5366191090467326
0.5366198039026356
0.5366204825923996
0.536621145673687
0.5366217936797877
0.5366224271194284
0.5366230464774651
0.5366236522183147
0.5366242447853409
0.5366248246045144
0.5366253920837349
0.5366259476111767
0.5366264915616991
0.5366270242922768
0.5366275461465295
0.5366280574558655
0.5366285585358422
         26002706 function calls in 15.908 seconds
```

```
   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000   15.908   15.908 {built-in method builtins.exec}
        1    0.000    0.000   15.908   15.908 nbody_opt.py:1(<module>)
        1    0.278    0.278   15.908   15.908 nbody_opt.py:132(nbody)
  2000000   14.854    0.000   15.627    0.000 nbody_opt.py:47(advance)
 20002000    0.640    0.000    0.640    0.000 {method 'add' of 'set' objects}
  4000401    0.133    0.000    0.133    0.000 {method 'keys' of 'dict' objects}
      100    0.002    0.000    0.002    0.000 {built-in method builtins.print}
      200    0.001    0.000    0.001    0.000 nbody_opt.py:86(report_energy)
        1    0.000    0.000    0.000    0.000 nbody_opt.py:114(offset_momentum)
        1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}
```

In [8]:

```
# profiling original nbody.py

!python -m cProfile -s cumulative assignments/hw1/nbody.py
```

-0.16908926275527172
-0.1690524049239141
-0.16901474404530115
-0.16902307738080738
-0.1690798593916716
-0.1690837851340391
-0.1690461237029509
-0.1690141632287673
-0.1690277860596755
-0.16908371256964286
-0.16908078067628346
-0.16904474158017624
-0.16901196403207439
-0.16903317301080864
-0.16908783999484456
-0.16908019793585344
-0.16904027123722912
-0.16900666759845165
-0.1690385994000913
-0.1690927820263521
-0.16907709741618646
-0.1690318157474469
-0.16900365578743826
-0.16904872400310705
-0.16909656666618936
-0.16907120739400114
-0.16902491372614678
-0.1690054028562737
-0.1690615084418881
-0.16909732952330989
-0.16906445026914277
-0.16901977601924423
-0.1690087837330612
-0.16906942996092264
-0.1690936981250834
-0.16905776550855287
-0.16901607122935539
-0.16901202092861942
-0.1690769590988439
-0.16909176966916228
-0.16905402040535616
-0.16901235909925366

-0.16901736033369455
-0.1690847459685499
-0.16909035534676262
-0.16904838452880422
-0.16900663034076266
-0.16902335367423574
-0.16909124149510837
-0.16908618459859356
-0.16904015517545168
-0.16900291923555205
-0.16903287494400335
-0.16909558331070956
-0.16908055117120765
-0.16903390308620184
-0.16900274590721814
-0.16904459300211724
-0.16909854774651836
-0.16907499697399825
-0.16902883385621698
-0.16900287426063337
-0.16905344993896332
-0.16909778695827116
-0.1690688542355601
-0.16902288517135491
-0.1690034111639226
-0.16906323202819348
-0.16909793327535813
-0.16906372791458069
-0.16901758963658736
-0.16900762655227408
-0.16907413362652127
-0.1690966701276139
-0.16905715253841647
-0.16901176533650106
-0.16901316541886924
-0.16908223578125162
-0.1690918842838697
-0.1690491567289713
-0.1690084483625472
-0.16902057135969534
-0.16908716660200046
-0.16908658831414558

```
-0.16904412116934314
-0.16900740673825185
-0.16902932050670785
-0.16909180276278374
-0.16908292959405574
-0.16903991544261052
-0.16900522540601676
-0.16903656773659115
-0.16909410318888132
-0.169078466211733
-0.16903274720873912
-0.16900284987920383
-0.1690462807510843
-0.1690971767866576
-0.16907293588692712
-0.16902628585294033
         326007706 function calls in 65.591 seconds

   Ordered by: cumulative time

   ncalls  tottime  percall  cumtime  percall filename:lineno(function)
        1    0.000    0.000   65.591   65.591 {built-in method builtins.exec}
        1    0.000    0.000   65.591   65.591 nbody.py:1(<module>)
        1    0.324    0.324   65.591   65.591 nbody.py:125(nbody)
  2000000   14.833    0.000   65.262    0.000 nbody.py:67(advance)
 20000000   16.598    0.000   45.620    0.000 nbody.py:54(update_vs)
120000000   12.966    0.000   29.022    0.000 nbody.py:47(compute_b)
120000000   16.056    0.000   16.056    0.000 nbody.py:51(compute_mag)
 10000000    1.760    0.000    1.760    0.000 nbody.py:62(update_rs)
 20002000    1.561    0.000    1.561    0.000 nbody.py:44(compute_deltas)
 20002000    0.878    0.000    0.878    0.000 {method 'append' of 'list' objects}
 14001401    0.610    0.000    0.610    0.000 {method 'keys' of 'dict' objects}
      100    0.002    0.000    0.002    0.000 {built-in method builtins.print}
      200    0.002    0.000    0.002    0.000 nbody.py:88(report_energy)
     2000    0.000    0.000    0.000    0.000 nbody.py:85(compute_energy)
        1    0.000    0.000    0.000    0.000 nbody.py:108(offset_momentum)
        1    0.000    0.000    0.000    0.000 {method 'disable' of '_lsprof.Profiler' objects}
```

Final Version of Optimization:

Original time: 65.591s Optimized time: 15.908s Speedup = 65.591 / 15.908 = 4.12